

Dataset before and after preprocessing

1- Data Cleaning:

1.1 - Duplicates: 34 duplicate rows are removed.

Before:

```
In [33]: df[df.duplicated()]
```

Out[33]:

	ID	Reason for absence	Month of absence	Day of the week	Seasons	Transportation expense	Distance from Residence to Work	Service time	Age	Work load Average/day	Disciplinary failure	Education	Son	Social drinker	Social smoker	Pet
16	3	23	7	6	1	179	51	18	38	239.554	0	1	0	1	0	0
68	34	23	10	3	4	118	10	10	37	253.465	0	1	0	0	0	0
88	28	23	11	4	4	225	26	9	28	306.345	0	1	1	0	0	2
109	10	22	12	4	4	361	52	3	28	261.306	0	1	1	1	0	4
127	34	27	1	2	2	118	10	10	37	308.593	0	1	0	0	0	0
128	34	27	1	3	2	118	10	10	37	308.593	0	1	0	0	0	0
131	34	27	1	4	2	118	10	10	37	308.593	0	1	0	0	0	0
132	27	23	1	5	2	184	42	7	27	308.593	0	1	0	0	0	0
133	34	27	1	5	2	118	10	10	37	308.593	0	1	0	0	0	0
305	5	23	10	2	4	235	20	13	43	265.017	0	1	1	1	0	0
351	3	28	12	6	4	179	51	18	38	236.629	0	1	0	1	0	0
376	3	27	2	4	2	179	51	18	38	251.818	0	1	0	1	0	0
385	3	27	2	4	2	179	51	18	38	251.818	0	1	0	1	0	0
386	3	27	2	6	2	179	51	18	38	251.818	0	1	0	1	0	0
388	3	27	2	4	2	179	51	18	38	251.818	0	1	0	1	0	0
389	3	27	2	6	2	179	51	18	38	251.818	0	1	0	1	0	0
440	22	23	5	4	3	179	26	9	30	246.074	0	3	0	0	0	0
477	24	28	7	3	1	246	25	16	41	230.290	0	1	0	1	0	0
496	24	28	9	3	1	246	25	16	41	261.756	0	1	0	1	0	0
560	28	23	12	4	4	225	26	9	28	280.549	0	1	1	0	0	2
605	3	27	2	4	2	179	51	18	38	264.249	0	1	0	1	0	0
607	3	27	2	6	2	179	51	18	38	264.249	0	1	0	1	0	0
610	3	27	2	2	2	179	51	18	38	264.249	0	1	0	1	0	0
615	3	27	2	4	2	179	51	18	38	264.249	0	1	0	1	0	0
616	3	27	2	5	2	179	51	18	38	264.249	0	1	0	1	0	0
631	3	27	3	4	2	179	51	18	38	222.196	0	1	0	1	0	0
632	3	27	3	5	2	179	51	18	38	222.196	0	1	0	1	0	0
641	3	27	3	4	2	179	51	18	38	222.196	0	1	0	1	0	0
643	3	27	3	5	2	179	51	18	38	222.196	0	1	0	1	0	0
666	22	27	4	6	3	179	26	9	30	246.288	0	3	0	0	0	0
666	22	27	4	6	3	179	26	9	30	246.288	0	3	0	0	0	0
669	22	27	4	6	3	179	26	9	30	246.288	0	3	0	0	0	0
673	22	27	4	6	3	179	26	9	30	246.288	0	3	0	0	0	0
699	15	28	5	5	3	291	31	12	40	237.656	0	1	1	1	0	1
700	22	27	5	6	3	179	26	9	30	237.656	0	3	0	0	0	0

34 rows × 21 columns

```
print(df.shape)

df.info()

print(df.isnull().sum())

print(df.duplicated().sum())

print(df.dtypes)

(740, 21)
```

After:

```
In [34]: #removing duplicates because repeated rows distort the analysis
```

```
df = df.drop_duplicates()
print("Shape after removing duplicates:", df.shape)
df[df.duplicated()]
```

Shape after removing duplicates: (706, 21)

Out[34]:

ID	Reason for absence	Month of absence	Day of the week	Seasons	Transportation expense	Distance from Residence to Work	Service time	Age	Work load Average/day	...	Disciplinary failure	Education	Son	Social drinker	Social smoker	Pet	Weight
0 rows × 21 columns																	

1.2 - Outliers: IQR used to remove outliers from these columns: distance_from_residence_to_work, age, work_load_average/day and absenteeism_time_in_hours.

```
Q1 = df[col].quantile(0.25)
Q3 = df[col].quantile(0.75)
IQR = Q3 - Q1

lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

outliers = df[(df[col] < lower_bound) | (df[col] > upper_bound)]

print(f"Outliers in {col}: {outliers.shape[0]}")

clean_df = df[(clean_df[col] >= lower_bound) & (clean_df[col] <= upper_bound)]

print("Original shape:", df.shape)
print("Clean shape:", clean_df.shape)

total_removed = df.shape[0] - clean_df.shape[0]
```

Outliers in distance_from_residence_to_work: 0
Outliers in age: 8
Outliers in work_load_average/day: 32
Outliers in absenteeism_time_in_hours: 44
Original shape: (706, 21)
Clean shape: (628, 21)

1.3 - Column Formatting: convert all column names to lowercase and replace spaces with underscores.

Before:

	ID	Reason for absence	Month of absence	Day of the week	Seasons	Transportation expense	Distance from Residence to Work	Service time	Age	Work load Average/day	...	Disciplinary failure	Education	Son	Social drinker	Social smoker	Pet	Weight
0	11	26	7	3	1	289	36	13	33	239.554	...	0	1	2	1	0	1	
1	36	0	7	3	1	118	13	18	50	239.554	...	1	1	1	1	0	0	

After:

	id	reason_for_absence	month_of_absence	day_of_the_week	seasons	transportation_expense	distance_from_residence_to_work	service_time	age	work
0	11		26	7	3	1	289	36	13	33
1	36		0	7	3	1	118	13	18	50

1.4 - Missing Values: No missing values.